

Acousticness

number [float]

A confidence measure from 0.0 to 1.0 of whether the track is acoustic. 1.0 represents high confidence the track is acoustic.

For all tracks: no preference

Danceability

number [float]

Danceability describes how suitable a track is for dancing based on a combination of musical elements including tempo, rhythm stability, beat strength, and overall regularity. A value of 0.0 is least danceable and 1.0 is most danceable.

Calm: High value – combo low value Tempo

Neutral: middle values in combo with middle values Tempo

Energy: Higher range – combo high value Tempo

Duration_ms

number [integer]

The duration of the track in milliseconds.

For all: set limits (not shorter than 1 min, not longer than 5 min)

Energy

number [float]

Energy is a measure from 0.0 to 1.0 and represents a perceptual measure of intensity and activity. Typically, energetic tracks feel fast, loud, and noisy. For example, death metal has high energy, while a Bach prelude scores low on the scale. Perceptual features contributing to this attribute include dynamic range, perceived loudness, timbre, onset rate, and general entropy.

Calm: low values

Neutral: middle values

Energy: higher range

Instrumentalness

number [float]

Predicts whether a track contains no vocals. "Ooh" and "aah" sounds are treated as instrumental in this context. Rap or spoken word tracks are clearly "vocal". The closer the instrumentalness value is to 1.0, the greater likelihood the track contains no vocal content. Values above 0.5 are intended to represent instrumental tracks, but confidence is higher as the value approaches 1.0.

For all tracks: no preference

Key

integer

The key the track is in. Integers map to pitches using standard [Pitch Class notation](#). E.g. 0 = C, 1 = C#/D♭, 2 = D, and so on. If no key was detected, the value is -1.

For all tracks: no preference

Liveness

number [float]

Detects the presence of an audience in the recording. Higher liveness values represent an increased probability that the track was performed live. A value above 0.8 provides strong likelihood that the track is live.

For all tracks: exclude live music – distraction/interaction of audio group feeling

Valence

number [float]

A measure from 0.0 to 1.0 describing the musical positiveness conveyed by a track. Tracks with high valence sound more positive (e.g. happy, cheerful, euphoric), while tracks with low valence sound more negative (e.g. sad, depressed, angry).

For all tracks: higher range

Tempo

number [float]

The overall estimated tempo of a track in beats per minute (BPM). In musical terminology, tempo is the speed or pace of a given piece and derives directly from the average beat duration.

For all tracks: look at danceability

Loudness

number [float]

The overall loudness of a track in decibels (dB). Loudness values are averaged across the entire track and are useful for comparing relative loudness of tracks. Loudness is the quality of a sound that is the primary psychological correlate of physical strength (amplitude). Values typically range between -60 and 0 db.

Calm: not too loud, set a bar and stick to it (equal loudness throughout playlist)

Neutral: mixed bag but within limits, smooth transitions

Energy: louder music, equal loudness throughout playlist (variance in song is encouraged)

Mode

integer

Mode indicates the modality (major or minor) of a track, the type of scale from which its melodic content is derived. Major is represented by 1 and minor is 0.

For all tracks: no preference

Speechiness

number [float]

Speechiness detects the presence of spoken words in a track. The more exclusively speech-like the recording (e.g. talk show, audio book, poetry), the closer to 1.0 the attribute value. Values above 0.66 describe tracks that are probably made entirely of spoken words. Values between 0.33 and 0.66 describe tracks that may contain both music and speech, either in sections or layered, including such cases as rap music. Values below 0.33 most likely represent music and other non-speech-like tracks.

For all tracks: no preference, exclude entirely spoken audiofiles (no music)

Time_signature

integer

An estimated time signature. The time signature (meter) is a notational convention to specify how many beats are in each bar (or measure). The time signature ranges from 3 to 7 indicating time signatures of "3/4", to "7/4".

For all tracks: no preference, depending on whether on-set not defined